Additive to Symptom Cluster Mapping – Research for Food Remedy API

This research document outlines an evidence-based mapping of food additives to specific symptom clusters, based on medical literature, allergen data, and regulatory guidelines (FSANZ, EFSA, UK FSA, etc.). The mapping focuses on four primary symptom categories: Respiratory, Gastrointestinal (GI), Dermatological, and Neurological. Each additive entry includes associated symptom clusters, evidence strength, sensitive groups, and regulatory notes. This document also suggests how to integrate these mappings into the Food Remedy API for user risk awareness and product exposure scoring.

Evidence-Based Mapping Table

INS / E Code	Additive (Class)	Symptom Cluster(s)	Sensitive Groups	Evidence Level	Regulatory / Notes
220–228	Sulphites (preservatives)	Respiratory (wheezing/asthma), Dermatological (hives), GI	Asthmatics, sulphite-sensitive	High	FSANZ: safe for most, but can trigger asthma in sensitive individuals
210–213	Benzoates (preservatives)	Dermatological (urticaria), Respiratory (asthma in aspirin-sensitive)	Aspirin-sensitive, chronic urticaria patients	Moderate	EFSA re-evaluation; links to urticaria/asthma documented
102, 104, 110, 122, 124, 129	Southampton 6 Colours	Neurological (hyperactivity), Dermatological (rashes)	Children	High	UK/EU require warning: 'may have adverse effect on activity and attention in children'
120	Carmine/cochineal (insect-	Dermatological (urticaria,	Carmine-	Moderate	Known allergen;

	derived colour)	anaphylaxis), Respiratory (rare)	allergic	–High	specific labelling in EU/US
160b	Annatto (natural colour)	Dermatological (urticaria/anaphylaxis), GI (nausea/vomiting)	Annatto-allergic	Moderate	Case reports in allergy literature
420, 421, 953, 965–968	Polyols (sweeteners)	GI (bloating, diarrhoea; laxative effect)	High consumers, IBS/FODMAP-sensitive	High	EU requires laxative warning if >10% polyols
249–250	Nitrates/nitrites	Respiratory/Blood (methemoglobinemia), Neurological (headache anecdotal)	Infants, high consumers	High (methemoglobinemia), Low (headache)	Strict usage limits (EU/FSANZ)
621, 620–625	MSG / glutamates	Neurological (headache, flushing), GI (nausea)	MSG-sensitive	Low–Moderate	FSANZ: safe for most; mild transient symptoms in small subset

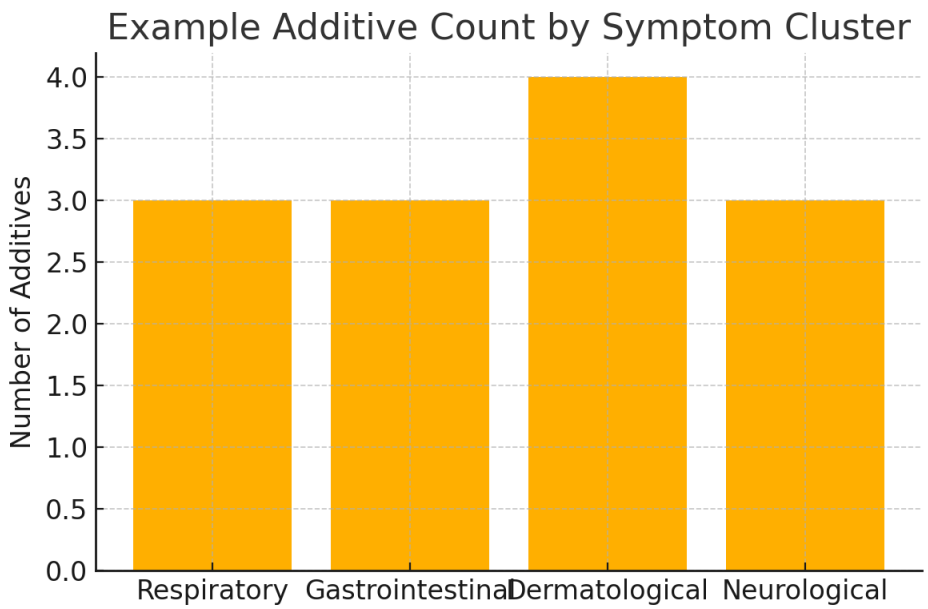
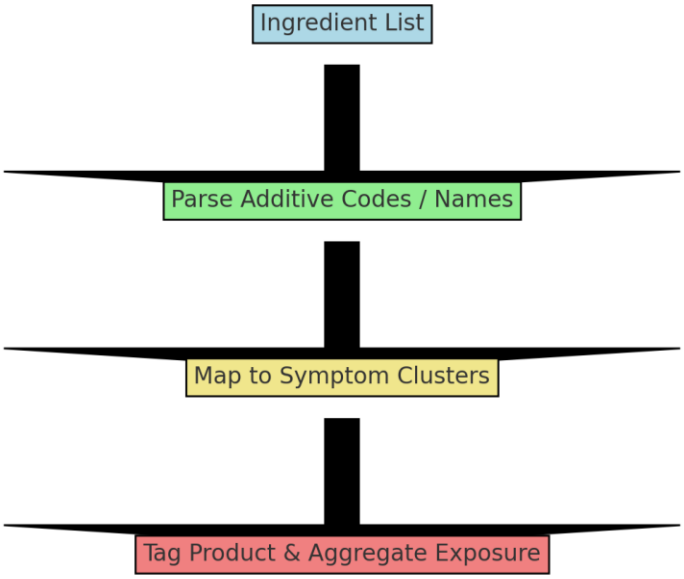
Methodology for Integration into Food Remedy API

1.Parse ingredient lists for additive names and INS/E codes.
2. Map matched additives to symptom clusters using this table.
3. Tag each product with the relevant clusters and evidence levels.
4. Aggregate exposure patterns across a user's basket to highlight potential risks.
5. Provide cluster badges (e.g., Respiratory, GI, Dermatological, Neurological) and tooltips with sensitive group warnings in the app.

Diagrams

Below are visual aids to accompany the additive mapping research.

1. Bar chart showing example additive counts by symptom cluster:



2. Process flowchart for mapping additives to symptom clusters in Food Remedy API: